

CSE 4631

Digital Signal Processing

Week 5

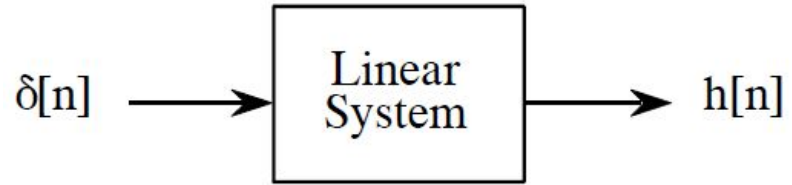
Recap (from Chapter 5)

- A signal can be decomposed into a group of components called **impulses** (Impulse decomposition).
- An impulse is a signal composed of all zeros except a single nonzero point, allowing to analyze signals one sample at a time.
- **Fundamental concept of DSP:** The input signal is decomposed into simple additive components, each of which is passed through a linear system, and the resulting output components are synthesized (added). The resulting signal is identical to that obtained by directly passing the *original signal* through the system.
- When impulse decomposition is used, the procedure can be described by a mathematical operation called **convolution**.

The Delta Function

- A *normalized* impulse, that is, sample number zero has a value of one, while all other samples have a value of zero.
- Also called the **unit impulse**.
- Symbolized by $\delta[n]$
- Any impulse can be represented as a *shifted* and *scaled* delta function.
- An impulse having the value of -3 in sample number 2, can be written as $a[n] = -3\delta[n-2]$

Impulse Response



- The signal that exits a system when a delta function (unit impulse) is the input.
- Differs for different systems.
- Usually symbolized by $h[n]$
- Because of **homogeneity** and **shift invariance**, if $\delta[n]$ results in $h[n]$, it follows that $-3\delta[n-2]$ results in $-3h[n-2]$
- If you know a system's impulse response, you immediately know how it will react to *any* impulse. Because the output will be a version of the impulse response that has been *shifted* and *scaled* by the same amount as the delta function on the input.

Convolution

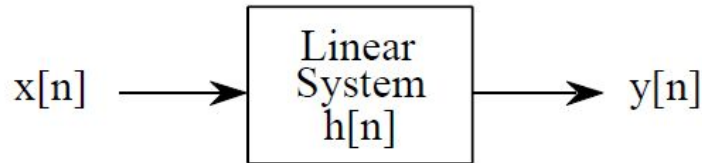
Let's summarize our way of understanding how a system changes an input signal into an output signal.

1. The input signal can be decomposed into a set of impulses, each of which can be viewed as a shifted and scaled delta function
2. The output resulting from each impulse is a scaled and shifted version of the impulse response
3. The overall output signal can be found by adding these scaled and shifted impulse responses.

Only thing to know: The system's impulse response.

Convolution (contd.)

- Convolution is a mathematical operation, which takes two *signals* and produces a third *signal*.
- Convolution is **commutative**: $a[n] \square b[n] = b[n] \square a[n]$
- Can be understood in two separate ways:
 - **Input side algorithm**: Analyzes how each sample in the input signal *contributes* to many points in the output signal.
 - **Output side algorithm**: Examines how each sample in the output signal has *received* information from many points in the input signal.



$$x[n] * h[n] = y[n]$$

Input Side and Output Side Algorithm

- For an input sequence of N samples and an impulse response of M samples, the output will have $N+M-1$ samples
- If the input samples are running from x_{min} to x_{max} and the impulse response is running from h_{min} to h_{max} , then the output will run from $(x_{min} + h_{min})$ to $(x_{max} + h_{max})$.

$$y[i] = \sum_{j=0}^{N-1} x[j] h[i-j]$$

$$y[i] = \sum_{j=0}^{M-1} h[j] x[i-j]$$

Equation: The **convolution summation**. This is the formal definition of convolution, written in the shorthand: $y[n] = x[n] \square h[n]$. In this equation, $h[n]$ is an M point signal with indexes running from 0 to $M-1$. And $x[n]$ is an N point signal with indexes running from 0 to $N-1$.

$$x[n] = \{-3, 4, \mathbf{1}, -2\}$$

$$h[n] = \{3, \mathbf{-2}, 1, 4\}$$

Find the output signal $y[n]$ using the input-side algorithm

Method 1: Solving equations

$$y[-3] = x[-2]h[-1] = -9$$

$$y[-2] = x[-2]h[0] + x[-1]h[-1] = 6 + 12 = 18$$

$$y[-1] = x[-2]h[1] + x[-1]h[0] + x[0]h[-1] = -3 - 8 + 3 = -8$$

$$y[0] = x[-2]h[2] + x[-1]h[-1] + x[0]h[0] + x[1]h[-1] = -12 + 4 - 2 - 6 = \mathbf{-16}$$

$$y[1] = x[-1]h[2] + x[0]h[1] + x[1]h[0] = 16 + 1 + 4 = 21$$

$$y[2] = x[0]h[2] + x[1]h[1] = 4 - 2 = 2$$

$$y[3] = x[1]h[2] = -8$$

Therefore, $y[n] = \{-9, 18, -8, \mathbf{-16}, 21, 2, -8\}$

Method 2: Convolution table

$x(n) \rightarrow$				-3	4	1	-2			
$y[-3] = -9$	4	1	-2	3						
$y[-2] = 6 + 12 = 18$		4	1	-2	3					
$y[-1] = -3 - 8 + 3 = -8$			4	1	-2	3				
$y[0] = -12 + 4 - 2 - 6 = -16$				4	1	-2	3			
$y[1] = 16 + 1 + 4 = 21$					4	1	-2	3		
$y[2] = 4 - 2 = 2$						4	1	-2	3	
$y[3] = -8$							4	1	-2	3

Therefore, $y(n) = \{-9, 18, -8, -16, 21, 2, -8\}$



Method 3: Convolution Matrix

$x(n) \rightarrow$ $h(n) \downarrow$	-3	4	1	-2
3	-9	12	3	-6
-2	-6	-8	-2	4
1	-3	4	1	-2
4	-12	16	4	-8

$$y(n) = \{-9, 18, -8, -16, 21, 2, -8\}$$